

Assignment for DAT246/DIT246

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CAUSAL ASSUMPTIONS & DAGs

The company is mainly interested in the effect of AI-assisted code review (used.cr.technology) on the extent and quality of code reviews. The binary decision variable result (whether the review contributed to making a decision) is conceptually downstream of extent and quality and is therefore not treated as a primary outcome variable in this analysis. So we consider the following variables: age, skill, used.cr.technology, complexity, extent, quality, result.

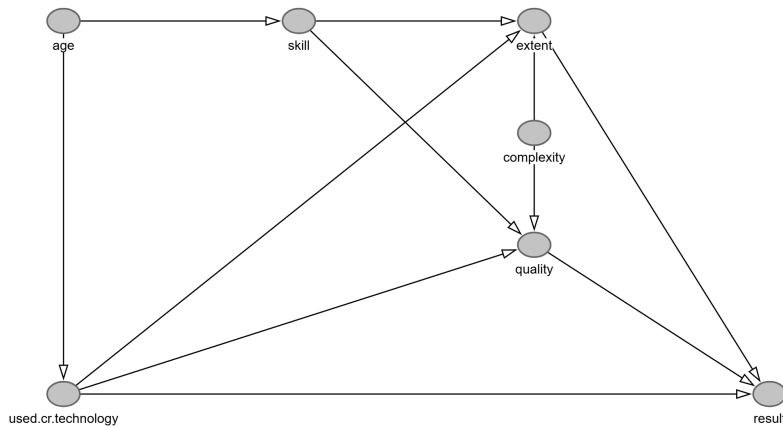


Figure 1: DAG 1 encodes the assumption that age influences skill and AI usage, while skill and complexity affect review extent and quality. AI usage directly affects both outcomes.

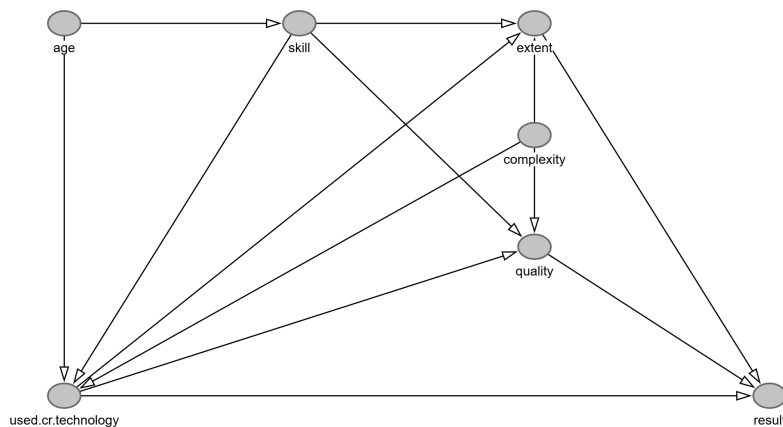


Figure 2: DAG 2 extends DAG 1 by allowing skill and complexity to influence AI adoption, introducing additional backdoor paths that require adjustment for age, skill, and complexity.

DESCRIPTIVE STATISTICS & VISUALS

Figure 3 shows the distribution of AI-assisted and non-AI code reviews. Both groups are well represented, indicating sufficient variation in the exposure variable (`used.cr.technology`) to study its effect on review `extent` and `quality`.

Figure 4 shows the distribution of review `extent` measured as the number of characters. The distribution is strictly positive and strongly right-skewed, with a long tail of very long reviews. This empirical shape supports modeling extent with a positive, skewed distribution rather than a Gaussian approximation.

Figure 5 displays the distribution of expert-assigned quality ratings on a 1–5 scale. The outcome is discrete and ordinal, with most reviews receiving mid-to-high scores. Treating quality as a continuous Gaussian variable would ignore its ordinal structure, motivating the use of an ordered logistic model.

STATISTICAL MODELS

Following option (a), we specify one statistical model per outcome variable. Review extent and review quality are the outcomes of interest.

Model for `extent`: Review extent is measured as the number of characters in a code review and is strictly positive. As shown by the descriptive statistics, the distribution of extent is right-skewed with a long tail. Under the maximum entropy principle, among all continuous distributions on $(0, \infty)$ with a fixed mean, the Exponential distribution has maximum entropy. We therefore model review extent using an Exponential generalized linear model with a log link. According to the DAGs, review extent is directly influenced by AI usage, reviewer age, reviewer skill, and code complexity. The model is specified as:

$$\text{extent}_i \sim \text{Exponential}(\lambda_i),$$

$$\log(\lambda_i) = \alpha_{\text{extent}} + \beta_{\text{extent, used.cr.technology}} \text{used.cr.technology}_i + \beta_{\text{extent, age}} \text{age}_i + \beta_{\text{extent, skill}} \text{skill}_i + \beta_{\text{extent, complexity}} \text{complexity}_i.$$

The log link ensures that the rate parameter λ_i is positive. All continuous predictors are standardized prior to modeling.

Model for `quality`: Review quality is assessed by experts on an ordinal scale from 1 (low quality) to 5 (high quality). Since the outcome is discrete and ordered, treating it as a continuous Gaussian variable would be inappropriate. We therefore model review quality using an ordered logistic regression.

$$\text{quality}_i \sim \text{OrderedLogit}(\phi_i, \kappa),$$

$$\phi_i = \alpha_{\text{quality}} + \beta_{\text{quality, used.cr.technology}} \text{used.cr.technology}_i + \beta_{\text{quality, age}} \text{age}_i + \beta_{\text{quality, skill}} \text{skill}_i + \beta_{\text{quality, complexity}} \text{complexity}_i.$$

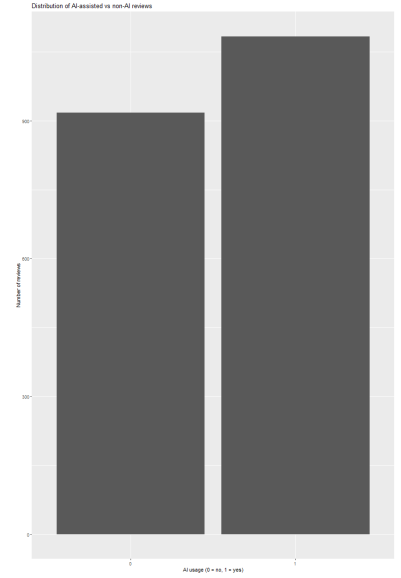


Figure 3: Distribution of AI Usage

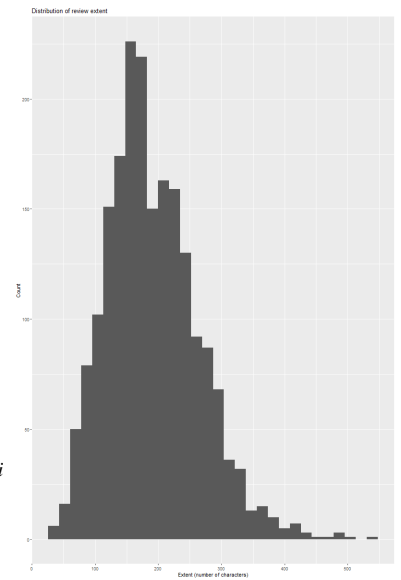


Figure 4: Distribution of extent

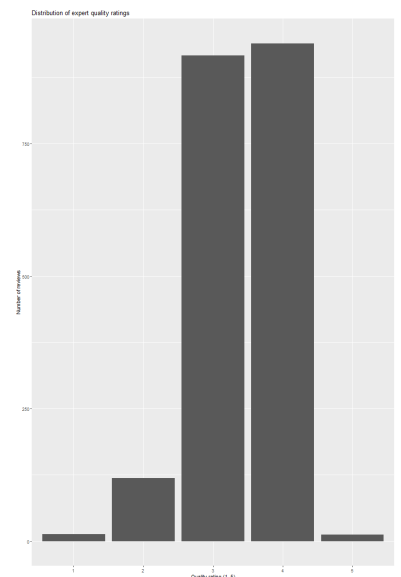


Figure 5: Distribution of quality

Here, $\kappa = (\kappa_1, \dots, \kappa_4)$ denotes the ordered cutpoints separating the five quality categories, and ϕ_i is the latent linear predictor. All continuous predictors are standardized prior to modeling.

SENSIBLE PRIORS

Before fitting any model to the observed data, we assess whether the chosen priors imply plausible data-generating processes. This step is particularly important for generalized linear models, where priors specified on transformed parameter scales (e.g., log or logit) can induce extreme or implausible behavior on the outcome scale.

Review extent is modeled with an exponential likelihood, and review quality with an ordered logistic likelihood, consistent with their empirical distributions and support.

Figure 6 compares the observed distribution of review extent with samples drawn from the prior predictive distribution. The prior predictive simulations generate strictly positive values with a right-skewed shape and a scale that overlaps well with the observed data. While the prior allows some larger values than commonly observed, these remain within a plausible range and do not dominate the probability mass. We therefore conclude that the priors for the extent model are weakly informative yet realistic.

Figure 7 compares the observed proportions of quality scores with the prior predictive distribution. The prior predictive samples cover all five categories and place non-zero probability mass across the full range. Although the prior distribution is flatter than the observed data — which is expected before conditioning on evidence — it does not concentrate probability on extreme or implausible categories. This indicates that the priors are conservative and do not encode strong assumptions about review quality before seeing the data.

Across both outcome variables, the prior predictive checks demonstrate that the chosen priors generate data consistent with the observed scale, support, and variability of the real data. The priors are therefore deemed eligible and suitable for posterior inference.

MODEL DIAGNOSTICS

Model training and convergence were assessed using trace plots, $\widehat{R}$, and effective sample sizes (ESS). Trace plots were inspected visually to verify that chains mix well and explore the same posterior regions. Quantitatively, $\widehat{R}$ values close to 1 indicate convergence, while large ESS values suggest low autocorrelation and stable estimation.

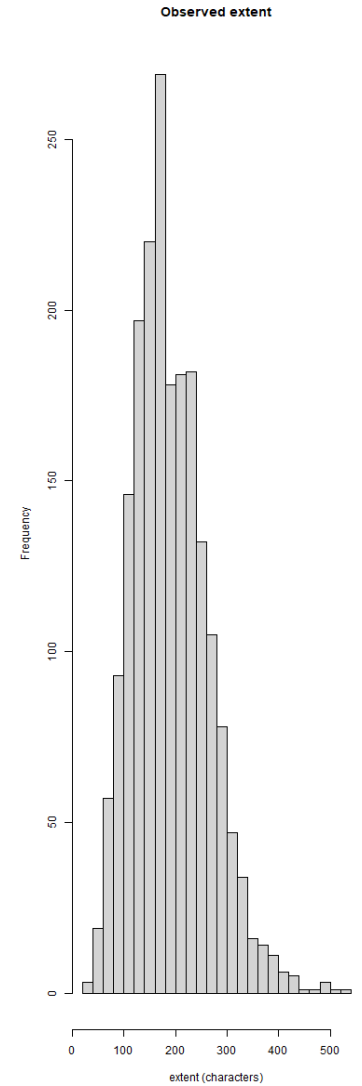


Figure 6: Observed and prior predictive distributions of review extent showing plausible scale and skewness.

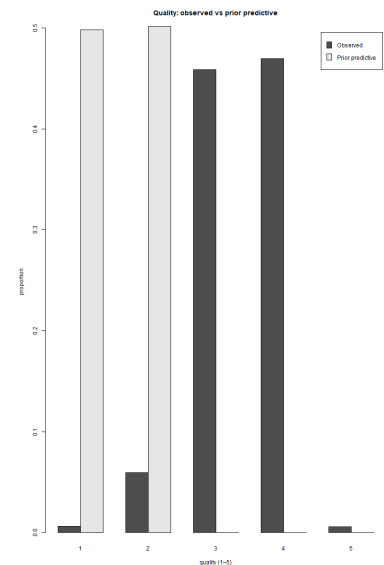


Figure 7: Observed and prior pre-

As shown in Figure 8, the extent model exhibits well-mixed chains for all regression coefficients, with no visible pathologies after warmup. All parameters have $\widehat{R} = 1.00$ and ESS values above 1,000, indicating efficient exploration of the posterior distribution.

Figure 9 shows that the quality model, including the ordered cutpoints, has well-mixed chains with substantial overlap across chains. All parameters have $\widehat{R} = 1.00$, and ESS values range up to approximately 2,300, demonstrating reliable convergence even for the more complex ordered logistic model.

Figure 10 provides a combined overview of the sampling behavior across parameters and confirms the absence of convergence issues.

Across both models, trace plots, $\widehat{R}$ values, and effective sample sizes consistently indicate successful model training. We therefore conclude that the posterior distributions obtained from these models can be used with confidence for inference and posterior prediction, conditional on the specified priors and likelihoods.

CONCLUSION

The goal of this analysis was to assess the causal effect of AI-assisted code review on review extent and review quality. Based on the causal assumptions encoded in the DAGs, review extent and review quality were treated as the primary outcomes of interest, while the binary decision variable `result` was considered a downstream consequence and not modeled directly.

Using generalized linear models aligned with the data-generating processes, we find clear evidence that AI usage is associated with higher review quality, while the effect on review extent appears small or near zero after adjusting for reviewer age, skill, and code complexity.

Model diagnostics indicate good convergence and effective sampling for all parameters, supporting the reliability of posterior inference. Prior predictive checks further confirm that the chosen priors are compatible with the observed data and do not impose implausible structure before seeing evidence.

Importantly, these conclusions are conditional on the assumed causal structure. Under DAG 1, adjusting for age, skill, and complexity blocks all backdoor paths between AI usage and the outcomes. Under DAG 2, where AI adoption is additionally influenced by reviewer skill and task complexity, adjusting for these variables remains sufficient to identify the causal effect. While unmeasured confounding cannot be ruled out, the results are consistent across plausible causal assumptions.

	mean	sd	5.5%	94.5%	rhat	ess_bulk
aE	-5.20	0.03	-5.26	-5.15	1.00	1077.75
bEC	-0.29	0.02	-0.32	-0.25	1.01	1475.31
bES	0.12	0.04	0.06	0.18	1.00	1076.76
bEA	0.00	0.04	-0.06	0.06	1.00	1054.42
bEU	-0.01	0.05	-0.08	0.07	1.00	1087.80

Figure 8: Trace plot extent.

	mean	sd	5.5%	94.5%	rhat	ess_bulk
aQ	0.41	0.59	-0.52	1.38	1.00	1354.78
bQC	-0.32	0.05	-0.39	-0.24	1.00	2253.57
bQS	1.17	0.07	1.05	1.29	1.00	1555.20
bQA	0.02	0.07	-0.10	0.15	1.00	1673.33
bQU	0.42	0.10	0.26	0.57	1.00	1888.72
cutpoints[1]	-5.22	0.63	-6.22	-4.20	1.01	1184.28
cutpoints[2]	-2.56	0.60	-3.50	-1.59	1.01	1330.16
cutpoints[3]	0.84	0.59	-0.08	1.83	1.00	1402.81
cutpoints[4]	6.08	0.63	5.09	7.10	1.01	1576.82

Figure 9: Trace plot quality.

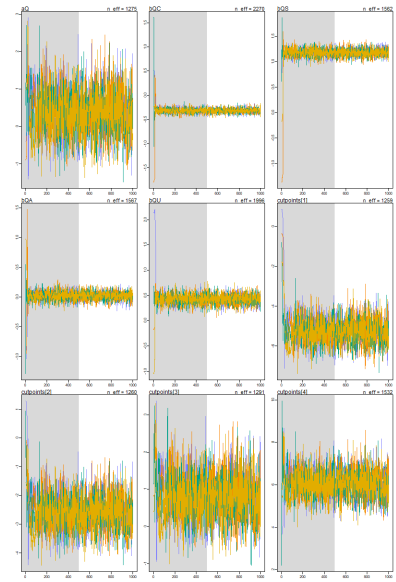


Figure 10: Diagnostics.